

Rohit Kandelkar,
Department of Computer Engineering (Regional Language),
PCCOE Pune, India,
Date: 30th August 2025

Subject: Resubmission of Manuscript ID - 157

Dear Editor,

We would like to thank you for reviewing the article Paper_ID- 157 and providing us with the reviewers' input and editorial evaluation. We have gone through the reviewers' comments and accordingly made changes to the manuscript. All changes are highlighted in yellow/ green/ blue color font in the marked copy of the manuscript. We kindly request you to review the manuscript mentioned in the subject for publication in your esteemed conference. Please find responses to reviewers' comments on the upcoming pages. The details of the marked copy of the manuscript are as follows:

The suggestions and the corresponding implementations from Reviewer 1 are highlighted in yellow, those from Reviewer 2 are highlighted in green, and those from Reviewer 3 are highlighted in blue.

Conference Name: ICATES 2025

Track Name: Artificial Intelligence, Machine Learning and Data Science

Manuscript ID: 157

Title: BLOOD GROUP DETECTION USING FINGERPRINT PATTERNS – A DEEP LEARNING APPROACH

All the Authors Full Name: Rohit Kandelkar, Yash Jahagirdar, Anandkumar Birajdar, Sukanya Bhaskar, and Rajeshwari Golande.

Sincerely,

Rohit Kandelkar

rohit.kandelkar22@pccoe pune.org

REVIEWER-I Suggestions & Implementation

Reviewer-I Suggestions & Implementation

1. *The reviewer recommended working on the paper's methodology and its presentation.*

Answer:

- The methodology section was expanded to include details on the dataset size, class distribution, and balancing approach.
- Standardized preprocessing steps, including undersampling, normalization, and augmentation, were added.
- A workflow diagram was included to visualize the methodology.
- The training configuration (optimizer, learning rate, early stopping, epochs) was clarified.
- The paper also implemented three additional models (Shallow CNN, Deep CNN, MLP) to allow for a better comparative analysis.

REVIEWER-II Suggestions & Implementation

Reviewer-II Suggestions & Implementation

- *The reviewer noted that the paper would benefit from a clearer discussion of its novelty compared to prior works, a more detailed dataset description, and additional experiments using baseline models for comparative validation.*

Answer:

- The abstract was revised to explicitly mention the dataset source (Kaggle), the multiple CNN architectures tested (Shallow, Custom, Deep), and the main findings, including the highest accuracy of 89.49% achieved by the Shallow CNN.

- The introduction was enhanced by adding the dataset source and expanding on the novelty of testing multiple CNN architectures.
- The literature review was improved with a critique of earlier CNN-based studies and a discussion of how this work uniquely compares different CNNs under identical preprocessing conditions.
- Three additional comparative models (Shallow CNN, Deep CNN, MLP) were implemented alongside the original Custom CNN.
- The results section was restructured to present a comparative performance analysis in tabular and graphical form, including confusion matrices.
- The conclusion was rewritten to reflect the comparative findings, highlighting the efficiency of the Shallow CNN and outlining future work.

REVIEWER-III Suggestions & Implementation

Reviewer-III Suggestions & Implementation

1. *This reviewer had a positive assessment of the paper's originality, technical soundness, significance, and results. No revisions were suggested by this reviewer.*